12 Probability: conditioning and weighted trees

Sheet 65 Tue 4 May 2027 *Probability model and equally likely outcomes*

1. Warm-up · Quick [PRB] A fair coin is tossed three times. How many outcomes are there?

   Ans. $2^3 = 8$ equally likely outcomes.

2. Core · Method [PRB] Biased die: $P(6) = 0.25$ and the other five faces are equally likely. Calculate $P(1)$.

   Ans. $\dfrac{1 - 0.25}{5} = 0.15$.

3. Core + · Critical thinking [PRB] "It will rain or it will not rain tomorrow: so there is a one in two chance." Is this equally-likely model justified?

   Ans. No: having two outcomes does not make them equally likely. Equal likelihood is an assumption of the model that must be justified.

4. Review · Quick [PER] Which shapes make up the net of a square-based pyramid?

   Ans. 1 square and 4 (isosceles) triangles.

5. Thread · Translate [LIT] Each side of a square of side x is increased by 3. Express the increase in area in terms of x.

   Ans. $(x+3)^2 - x^2 = 6x + 9$.

Sheet 66 Mon 10 May 2027 *Law of large numbers (informal version): simulation*

1. Warm-up · Quick [STA] In 100 tosses, 47 are "heads". What is the relative frequency of "heads"?

   Ans. 0.47.

2. Core · Interpret [PRB] Why, in 10 tosses of a fair coin, can the frequency of "heads" stray from 0.5? What happens with 10,000 tosses?

   Ans. Sampling fluctuation: a small sample has high variability. For large samples, the frequency stabilises near the probability 0.5 (law of large numbers).

3. Core + · True/false [PRB] True or false? Justify. Tossing a fair coin 1,000 times gives exactly 500 "heads".

   Ans. False. The number obtained is close to 500, rarely exactly 500.

4. Review · Problem [MOD] Price including tax $P(x) = 1.2\,x$ (20% VAT). Calculate $P(250)$ and the preimage of 480.

   Ans. $P(250) = 300$; $1.2x = 480 \Leftrightarrow x = 400$.

5. Thread · Method [EQ] Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

   Ans. $x = 14$

   Cross-multiplication: $2(2x-1) = 3(x+4)$.

Sheet 67 Tue 11 May 2027 *Python simulation of a repeated random experiment*

1. Warm-up · Quick [PRB] $P(A) = 0.35$. Calculate $P(\overline{A})$.

   Ans. $1 - 0.35 = 0.65$.

2. Core · Reading code [ALG] What does this program compute? Roughly what value will it display?

```
from random import randint
compte = 0
for i in range(1000):
    if randint(1, 6) == 6:
        compte = compte + 1
print(compte / 1000)
```

   Ans. The frequency of a 6 in 1,000 rolls of a die; about $\dfrac{1}{6} \approx 0.167$.

3. Core + · Method [PRB] Fair 12-sided die with faces numbered 1 to 12. Probability of getting a prime

number?

Ans. Primes: 2 ; 3 ; 5 ; 7 ; 11: $\frac{5}{12}$ (1 is not prime).

4. Review · Construction [CON] M lies on the perpendicular bisector of $[AB]$. What can be said about MA and MB? How is it constructed with compasses?

   Ans. $MA = MB$. Construction: two arcs of the same radius (greater than $\frac{1}{2}AB$) centred at A and B; their intersections define the perpendicular bisector.

5. Thread · Recognise [SOL] What is the shape of the cross-section of a cube by a plane parallel to one of its faces?

   Ans. A square with the same side as the face.

Sheet 68 Thu 13 May 2027 *Conditional probability $P_A(B)$*

1. Warm-up · Quick [PRB] Among 25 canteen users, 15 are girls. A canteen user is chosen at random: probability that this is a girl?

   Ans. $\frac{15}{25} = 0.6$.

2. Core · Method [PRB] In a class of 30 pupils, 12 play football, of whom 5 also play handball. A pupil is chosen at random: calculate $P_F(H)$, where F: "plays football", H: "plays handball".

   Ans. $P_F(H) = \frac{\text{Card}(F \cap H)}{\text{Card}(F)} = \frac{5}{12}$.

3. Core + · Method [PRB] In an equally likely model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.

   Ans. $\frac{20}{50} = 0.4$.

4. Review · Quick [NUM] Put in increasing order: $\frac{7}{4}$, $-\frac{3}{2}$, 0.5.

   Ans. $-\frac{3}{2} < 0.5 < \frac{7}{4}$ ($-1.5 < 0.5 < 1.75$).

5. Thread · True/false [DEM] True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

   Ans. False. Counter-example: a trapezium (not a parallelogram) has only one pair of parallel sides.

Sheet 69 Tue 18 May 2027 *Conditional probability and two-way tables*

1. Warm-up · Quick [PRB] In the following table, what is the total of the "sport" row?

	Girls	Boys	Total
Sport	18	22	…
No sport	12	8	20
Total	30	30	60

   Ans. $18 + 22 = 40$.

2. Core · Method [PRB] A pupil is chosen at random from this table. Calculate $P(S)$, $P_F(S)$ and $P_S(F)$ (F: girl, S: does sport).

   Ans. $P(S) = \frac{40}{60} = \frac{2}{3}$; $P_F(S) = \frac{18}{30} = 0.6$; $P_S(F) = \frac{18}{40} = 0.45$.

3. Core + · True/false [PRB] True or false? Justify. In this table, $P_F(S) = P_S(F)$.

   Ans. False. $0.6 \neq 0.45$: the reference sets (30 girls and 40 sports participants) differ.

4. Review · Quick [PUI] Calculate $5^0 + 2^{-1}$.

   Ans. $1 + \frac{1}{2} = \frac{3}{2}$.

5. Thread · Method [REP] $ABCD$ is a parallelogram with $A(1,1)$, $B(5,2)$, $C(6,5)$. Find D.

Ans. The diagonals have the same midpoint: $D = A + C - B = (2, 4)$. Check: $\overrightarrow{AB} = \overrightarrow{DC} = (4, 1)$.

Sheet 70 Thu 20 May 2027 *Weighted trees: construction*

1. Warm-up · Quick [PRB] What is the sum of the probabilities of the branches leaving a single node?
   Ans. 1.
2. Core · Method [PRB] In a population, 30% have a disease M. A test is positive (T) for 90% of those with the disease and for 5% of those without it. Complete the branch probabilities of the tree.
   Ans. $P(M) = 0.3$, $P(\overline{M}) = 0.7$; $P_M(T) = 0.9$, $P_M(\overline{T}) = 0.1$; $P_{\overline{M}}(T) = 0.05$, $P_{\overline{M}}(\overline{T}) = 0.95$.
3. Core + · Translate [PRB] Translate into symbols: "among those with the disease, 90% test positive".
   Ans. $P_M(T) = 0.9$: branch $M \to T$; do not write $P(T) = 0.9$.
4. Review · Quick [CAL] Calculate $\dfrac{2}{3} \div \dfrac{4}{9}$.
   Ans. $\dfrac{2}{3} \times \dfrac{9}{4} = \dfrac{18}{12} = \dfrac{3}{2}$.
5. Thread · Problem [DIV] Three lighthouses flash every 6, 8 and 9 seconds. They light up together at $t = 0$. When is the next simultaneous flash?
   Ans. $\text{lcm}(6, 8, 9) = 72$: after 72 s.

Sheet 71 Mon 24 May 2027 *Weighted trees: multiplying along a path*

1. Warm-up · Quick [CAL] Calculate 0.3×0.9.
   Ans. 0.27.
2. Core · Method [PRB] Using the tree from sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.
   Ans. $P(M \cap T) = 0.3 \times 0.9 = 0.27$; $P(\overline{M} \cap T) = 0.7 \times 0.05 = 0.035$.
   We multiply along a path: $P(M \cap T) = P(M) \times P_M(T)$.
3. Core + · Method [PRB] Deduce $P(T)$.
   Ans. Two paths lead to T: $0.27 + 0.035 = 0.305$.
4. Review · Factorise [DEV] Factorise $x^2 - 9$.
   Ans. $(x - 3)(x + 3)$.
5. Thread · Quick [GEO] The side $[BC]$ of length 13 cm is the hypotenuse of a right-angled triangle ABC. What is the radius of its circumscribed circle?
   Ans. The midpoint of the hypotenuse is the centre: radius $= 13 \div 2 = 6.5$ cm.

Sheet 72 Tue 25 May 2027 *From table to tree and symbolic notation*

1. Warm-up · Quick [PRB] Express $\dfrac{30}{60}$ as a probability.
   Ans. 0.5.
2. Core · Method [PRB] From the table on sheet 69, build the tree (F or G, then S or $\overline{S}$): give $P(F)$, $P_F(S)$ and $P_G(S)$.
   Ans. $P(F) = 0.5$; $P_F(S) = \dfrac{18}{30} = 0.6$; $P_G(S) = \dfrac{22}{30} = \dfrac{11}{15}$.
3. Core + · Translate [PRB] Write symbolically and calculate "the probability of being a boy who does not do sport".
   Ans. $P(G \cap \overline{S}) = \dfrac{8}{60} = \dfrac{2}{15}$; or $0.5 \times \dfrac{8}{30}$.
4. Review · Quick [QP] A car travels at constant speed: 150 km in 2 h. What distance in 5 h?
   Ans. $75 \times 5 = 375$ km.

5. Thread · Statement → formula [REG] A taxi charges a pick-up fee of 2,000 Ar then 1,500 Ar per km. Express the price $p(x)$ for x km, then calculate $p(8)$.

   Ans. $p(x) = 1{,}500x + 2{,}000$; $p(8) = 14{,}000$ Ar.

Sheet 73 Thu 27 May 2027 *Reversing conditioning: $P_A(B)$ and $P_B(A)$*

1. Warm-up · Quick [PRB] What is the difference between $P_M(T)$ and $P_T(M)$?

   Ans. $P_M(T)$: probability of testing positive given that one has the disease; $P_T(M)$: of having the disease given that one tests positive.

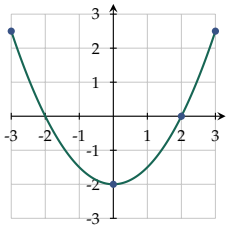
2. Core · Method [PRB] Using the data from sheet 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.

   Ans. $P_T(M) = \dfrac{P(M \cap T)}{P(T)} = \dfrac{0.27}{0.305} \approx 0.885$.

3. Core + · Method [PRB] Rare disease (1% of the population); the test is positive for 99% of those with the disease and 5% of those without. Calculate $P_T(M)$.

   Ans. $P(M \cap T) = 0.0099$; $P(\overline{M} \cap T) = 0.99 \times 0.05 = 0.0495$; $P(T) = 0.0594$; $P_T(M) = \dfrac{0.0099}{0.0594} = \dfrac{1}{6} \approx 0.17$.
   Counter-intuitive result: a test that is "99% reliable" is not enough.

4. Review · Graph [IMG] Curve of f:

   Read $f(2)$ and the preimages of 2.5.

   Ans. $f(2) = 0$; preimages of 2.5: -3 and 3.
   Expression used for the plot: $f(x) = 0.5x^2 - 2$.

5. Thread · Method [ALG] Program: choose a number; multiply it by -2; add 10; divide by 2. Result for 7? Expression in x?

   Ans. $\dfrac{-14 + 10}{2} = -2$; in general $\dfrac{-2x + 10}{2} = -x + 5$.

Sheet 74 Mon 31 May 2027 *False positives and false negatives*

1. Warm-up · Quick [PRB] Define "false positive" and "false negative".

   Ans. False positive: positive test for a person who does not have the disease. False negative: negative test for a person who has the disease.

2. Core · Method [PRB] 10,000 people, 2% with the disease; the test is positive for 90% of those with the disease and 10% of those without. How many false negatives, false positives and positives in total?

   Ans. With the disease 200: 180 positive, so 20 false negatives; without the disease 9,800: 980 false positives; positives: $180 + 980 = 1{,}160$.

3. Core + · Interpret [PRB] Using the previous data, calculate the proportion of people who truly have the disease among the positives and comment.

   Ans. $\dfrac{180}{1{,}160} \approx 15.5\%$: since the disease is rare, most positives are false positives.

4. Review · True/false [PNP] True or false? Justify. The area of a square is proportional to the length of its side.

   Ans. False. $c = 1 \mapsto 1$ and $c = 2 \mapsto 4 \neq 2 \times 1$.

5. Thread · Disprove [LIT] Conjecture: "for every integer n, $n^2 + n + 41$ is prime". Test $n = 40$.

Ans. $40^2 + 40 + 41 = 1681 = 41^2$: not prime. The conjecture is false.

Successful tests prove nothing; one failed test disproves.